

ARIANNA FERNANDEZ

Boston, MA | arianna.fernandez.fermin@gmail.com | 786-448-9798 | [Arianna Alexandra Fernandez Fermin](#) | [LinkedIn](#)

EDUCATION

UNIVERSITY OF MASSACHUSETTS BOSTON

Bachelor of Science in Computer Science; Minor in Economics
GPA: 4.00 | Honors College | Alpha Lambda Delta

Boston, MA
Expected May 2028

RESEARCH EXPERIENCE

MACHINE LEARNING FAIRNESS RESEARCH

Researcher – Evaluating

Boston, MA
Jan 2026 - Present

- Developing an interpretable ML credit-scoring model to evaluate socioeconomic bias, proxy-variable effects, and threshold-based approval outcomes.
- Built Python pipelines to compare approval scenarios using disparate impact, demographic parity, equal opportunity, calibration, and predictive equality metrics: tested with different profiles, conducting thematic analysis on variables.
- Designed transparent scoring logic with interaction weights, uncertainty bounds, and conservative decision overlays for explainable lending analysis.

IEEE ETHICS CONFERENCE PUBLICATION

First Author Researcher

Boston, MA / Chicago, IL
Oct 2024 - Jun 2025

- Published and presented first-author research on belonging and retention of Black women and Latinas in Computer Science at an urban public university.
- Applied mixed-methods analysis using Python/Pandas for quantitative exploration and thematic coding for qualitative findings.

THINKNEURO SUMMER RESEARCH PROGRAM

Research Intern - BCI Bibliometrics

Online
Jun 2025 - Aug 2025

- Processed scientific records from Web of Science and performed bibliometric analysis on brain-computer interface technologies for motor rehabilitation.
- Used R for data cleaning, exploratory analysis, trend visualization, and graduate-level poster/abstract development.

EXPERIENCE

AI/ML DOCUMENT INTELLIGENCE EXTERNSHIP

AI/ML Extern

Remote
Jan 2026 - Present

- Built OCR/NLP pipelines in Python using PyMuPDF and pdfplumber to convert unstructured mortgage documents into searchable datasets.
- Prototyped RAG systems with LlamaIndex and FAISS and evaluated open-source LLM performance across document understanding tasks.

UNIVERSITY OF MASSACHUSETTS BOSTON

Supplemental Instruction Leader - Data Structures & Algorithms-Part-Time Job

Boston, MA
Sep 2025 -Present

- Lead weekly Java sessions on data structures, debugging, OOP design, searching/sorting, hashing, trees, and time/space complexity.

NATIVE AND INDIGENOUS STUDENT ASSOCIATION

Boston, MA

Community Leader - Part-Time Job

Feb 2025 - May 2026

- Coordinate cultural and educational programming, budget logistics, scheduling, outreach, and student communications for campus events.

LEADERSHIP & PROJECTS

CS RESEARCH HUB

Student Manager

Boston, MA
2026 - Present

- Support undergraduate CS research initiatives, student engagement, research communication, and event/program coordination.

WORLD ROBOT OLYMPIAD

Future Innovator - National Winner / International Finalist

Venezuela / Panama
2023

- Led C++/Arduino hardware-software prototype addressing low-connectivity sales and safety challenges for fishing communities.

SKILLS

Programming: Python, Java, C++, C, R, SQL | **Frontend:** HTML, CSS, React | **AI/ML/Data:** Pandas, NumPy, OCR, NLP, RAG, LlamaIndex, FAISS, fairness metrics, EDA, visualization | **Tools:** Git, Excel, PowerPoint, Microsoft 365 | **Languages:** Spanish native, English fluent, Italian basics